Akoizo Original Practice - Form 3

Division B | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Parallel resistors share…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. Voltage

B. Always current

C. Always power

D. Always resistance

Answer: _____

2. (2 points) KCL expresses…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Charge conservation

B. Equal current in all branches

C. Energy creation

D. Resistance addition only

Answer: _____

3. (2 points) Which change best tests the cause of the dark LED?

Unit: Fault finding with switches, fuses, and diodes.
Worked-case data: Supply: The battery reads 6 V. Branch A: Its LED lights through a resistor. Branch B: Its LED is reversed and remains dark.
Separate investigation observations:
Supply: The battery reads 6 V.
Branch A: Its LED lights through a resistor.
Branch B: Its LED is reversed and remains dark.

A. Remove both current-limiting resistors

B. Reverse the LED with power disconnected, retaining its resistor

C. Short the battery terminals

D. Replace the working battery without checking wiring

Answer: _____

4. (2 points) Two 100 Ω resistors in series give…

A. 200 Ω

B. 50 Ω

C. 100 Ω

D. 10000 Ω

Answer: _____

5. (2 points) Negative solved current means…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Opposite the assumed direction

B. Impossible circuit necessarily

C. Negative charge disappears

D. No voltage exists

Answer: _____

6. (4 points) Explain the evidence supporting your answer, including the calculation or comparison.

Unit: Fault finding with switches, fuses, and diodes.
Worked-case data: Supply: The battery reads 6 V. Branch A: Its LED lights through a resistor. Branch B: Its LED is reversed and remains dark.
Separate investigation observations:
Supply: The battery reads 6 V.
Branch A: Its LED lights through a resistor.
Branch B: Its LED is reversed and remains dark.

Answer: _____
_____
_____

7. (2 points) An open switch can have…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. Voltage with zero current

B. Infinite real power

C. Always zero voltage

D. Consumed charge

Answer: _____

8. (2 points) Resistor dissipation is…

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. I^2R

B. Always negative

C. Independent of current

D. Equal to charge

Answer: _____

9. (4 points) Explain "Open circuit" in your own words.

Unit: Fault finding with switches, fuses, and diodes.
Worked-case data: Supply: The battery reads 6 V. Branch A: Its LED lights through a resistor. Branch B: Its LED is reversed and remains dark.
Separate investigation observations:
Supply: The battery reads 6 V.
Branch A: Its LED lights through a resistor.
Branch B: Its LED is reversed and remains dark.

Answer: _____
_____
_____

10. (4 points) Why trace nodes first?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

Answer: _____
_____
_____

11. (4 points) Write a source-two-resistor loop equation.

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: Supply: The battery reads 6 V. Branch A: Its LED lights through a resistor. Branch B: Its LED is reversed and remains dark.

Answer: _____
_____
_____

13. (4 points) Find 100 Ω parallel 100 Ω.

Answer: _____
_____
_____

14. (4 points) Why avoid dependent equations?

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

Answer: _____
_____
_____

15. (4 points) Name a useful reference feature.

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

Answer: _____
_____
_____

16. (4 points) What does a relay separate?

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

Answer: _____
_____
_____

17. (4 points) Why isolate an unknown for resistance measurement?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

Answer: _____
_____
_____

18. (4 points) Does zero field imply zero potential?

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

Answer: _____
_____
_____